A second circuit design, shown in Figure 4, is chosen as the 6.0-V DC power supply.

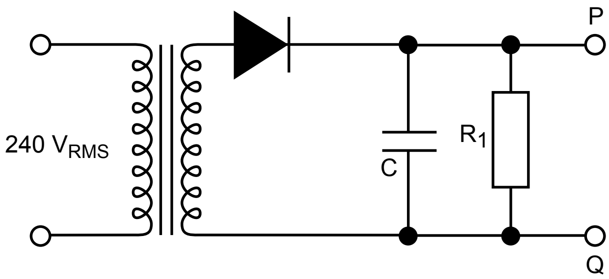

Figure 4

Question 8

What is the purpose of the resistor-capacitor (R_1 – C) combination in Figure 4?

2 marks

Question 9

The time constant of the resistor-capacitor combination for the circuit in Figure 4 is 2.0 s. If C = 1000 μF, what is the value of R_1?

R_1 = Ω

2 marks

To check the performance of the 6-V DC power supply, the voltage ripple was measured between points P and Q of Figure 4. This signal was amplified using an amplifier with a voltage gain of +100. **The amplified voltage** was displayed on an oscilloscope, as shown in Figure 5.

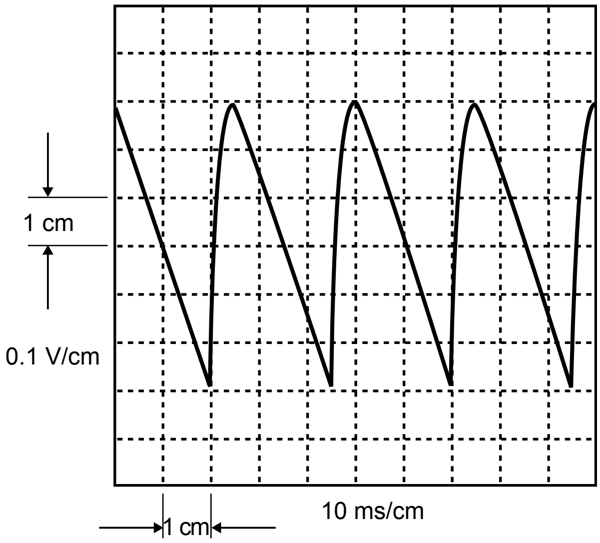

Figure 5

Question 10

From Figure 5, what is the peak-to-peak value of the voltage ripple of the signal between points P and Q? Give your answer in units of mV.

$V_{\text{peak to peak}}$ = mV

3 marks

The voltage across the air-temperature sensor (V_{AIR}), and that across the water-temperature sensor (V_{WATER}), are the inputs to a digital electronic system that controls the water pump for the solar panel. The logic levels of these input digital signals are

air temperature	logic level, L_{AIR}
<20°C	0
≥20°C	1

water temperature	logic level, L_{WATER}
>30°C	0
≤30°C	1

The output of the digital circuit is the pump-control signal, L_{PUMP}. If this is logic 0 the pump is OFF (water does not circulate through the solar heater), if this is logic level 1 the pump is ON.

Question 11

Complete the last two rows of the truth table for this digital system.

L_{AIR}	L_{WATER}	L_{PUMP}
0	0	0
0	1	0
1	0	
1	1	

2 marks

Question 12

Which digital circuit **(A–D)** in Figure 6 will correctly implement your completed truth table from Question 11?

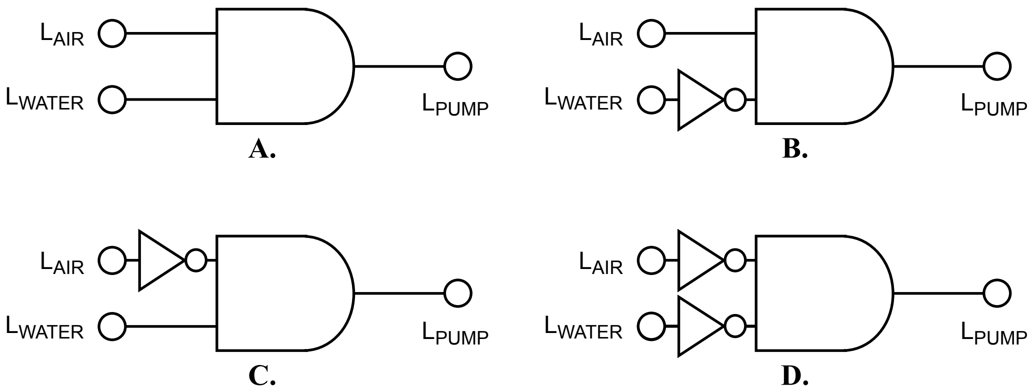

Figure 6

2 marks

The air-temperature-sensing circuit requires a light-emitting diode (LED) to indicate if the air-temperature is above 20˚C (LED ON) or below 20˚C (LED OFF). The circuit used is shown in Figure 7a, and the non-linear I-V characteristic of the LED is shown in Figure 7b.

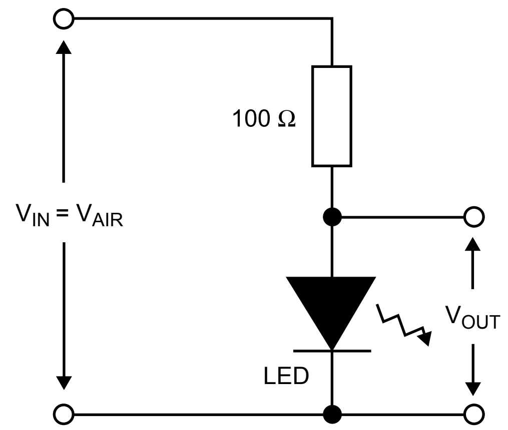

Figure 7a

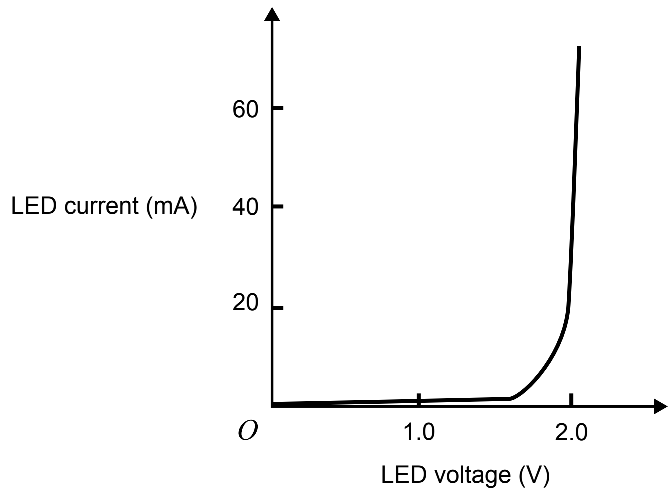

Figure 7b

Question 13

What is the current in the circuit of Figure 7a when V_{AIR} = 4.5 V? Show your working, and give your answer in units of mA.

Current = mA

3 marks

END OF QUESTION AND ANSWER BOOK